Getting Started Guide
Four Topology MOSbius

Prerequisites: Download the GitHub Desktop, Docker Desktop, and TigerVNC

Open GitHub Desktop, and clone our team repository. Place it wherever on your computer and wait for it to download.

Now, make sure that the Docker Desktop app is running (just make sure that it is opened on the container page, you shouldn't have any containers right now). Open your terminal and navigate to wherever you cloned our team repository. Once you are in the folder, run the launch script, which is ./start_chipathon_vnc.sh on Mac, and .\start_chipathon_vnc.bat on Windows. Docker will pull the container image, and it will take a few minutes to finish.

Now, you should see the container in Docker. Make sure that it is running (click to play button), and then open TigerVNC. In the connection box, type localhost:5901 as the server address and click connect. You will be prompted to put in the password, it is abc123.

Now that you are connected to the Linux desktop, open the terminal. It should automatically put you in the correct environment (the GF PDK). Directly from opening the terminal, you can launch xschem by typing in "xschem". You can open any existing schematics and create any new schematics and testbenches you want.

The folder at /foss/designs/ inside the container is the same folder as the designs folder in your cloned repository on your host computer.

Now that the environment is set up, review the team workflow.

Team Workflow on GitHub

All work happens on feature branches, never directly on main. This lets us work in parallel without stepping on each other and gives every change a review before it joins the shared codebase.

Before creating a branch, make sure your local main is current. In GitHub Desktop, switch to the main branch using the top-left branch dropdown, then click Fetch origin and Pull origin if there are changes to pull. Skipping this step means your new branch starts from an outdated version of the project, which can create conflicts when you eventually merge.

To create your branch, click the current branch dropdown at the top of GitHub Desktop, then click New Branch. Name it "feature/" followed by what you are working on. For example, if you wanted to work on the Two-Stage OTA, you could name it "feature/two-stage." After creating the branch, click Publish branch to push the empty branch up to GitHub. This protects your work in case of a laptop issue and confirms everything is set up correctly.

Once your branch is published, do your design work in Xschem (or whichever tool you are using). Save your files normally. All saves go into the repo folder automatically because of how the container is mounted to your host machine. When you have finished a meaningful amount of work, such as a working schematic, a successful simulation, or a completed layout pass, come back to GitHub Desktop. You will see your changed files listed in the left pane. Write a short, descriptive commit summary. Click Commit to feature/your-branch, and then click Push origin to send your commits to GitHub. Commit often, because each commit is a checkpoint you can revert to if something breaks.

When your work is ready for the team to review, which does not necessarily mean perfect but rather a meaningful checkpoint, open a pull request on GitHub.com. Go to the repo on GitHub and you will see a banner that says "Compare & pull request." Click it, write a description of what you built and how you verified it, and request a teammate as the reviewer. Once approved, the reviewer or the team lead clicks Merge pull request on GitHub, which brings your work into main. After merging, click Delete branch on GitHub to remove the finished branch from the remote. After the pull request is merged, you need to clean up locally. In GitHub Desktop, switch back to the main branch, then click Fetch origin and Pull origin to bring your merged work onto your local main. Delete the local feature branch by clicking the branch dropdown, right-clicking your feature branch, and selecting Delete. Now you are back on a current main with a clean branch list, ready to start the next task.